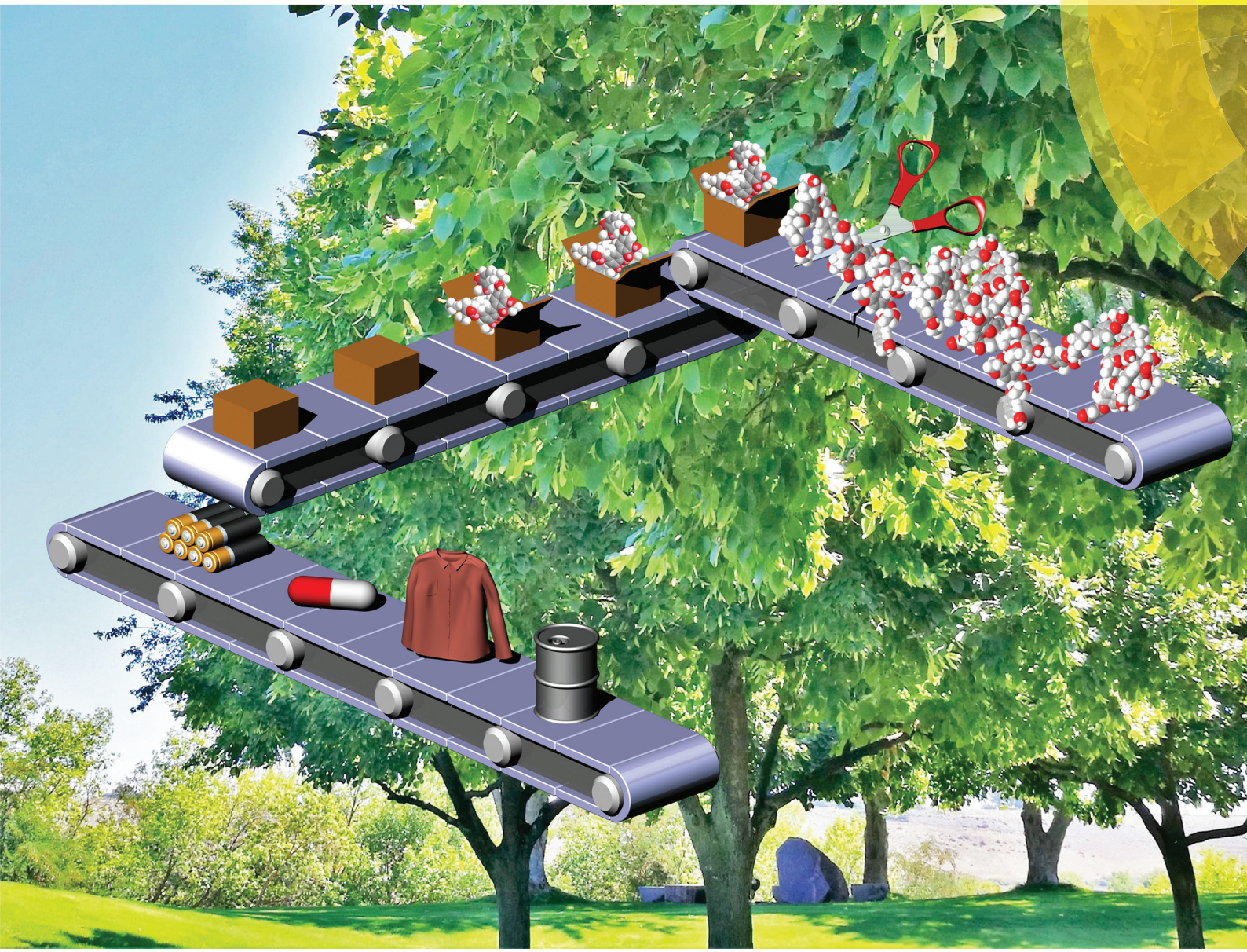
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Xiao Zhang *et al.*

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Unique low-molecular-weight lignin with high purity extracted from wood by deep eutectic solvents (DES): a source of lignin for valorization†

Carlos Alvarez-Vasco,‡[a] Ruoshui Ma,‡[a] Melissa Quintero,[a] Mond Guo,[a] Scott Geleynse,[a] Karthikeyan K. Ramasamy,[b] Michael Wolcott[c] and Xiao Zhang*[a]

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This paper reports a new method of applying Deep Eutectic Solvents (DES) for extracting lignin from woody biomass with high yield and high purity. DES mixtures prepared from choline chloride (ChCl) and four hydrogen-bond donors – acetic acid, lactic acid, levulinic acid and glycerol – were evaluated for treatment of hardwood (poplar) and softwood (D. fir). It was found that these DES treatments can selectively extract a significant amount of lignin from wood with high yields: 78% from poplar and 58% from D. fir. The extracted lignin has high purity (95%) with unique structural properties. We discover that DES can selectively cleave ether linkages in wood lignin and facilitate lignin removal from wood. The mechanism of DES cleavage of ether bonds between phenylpropane units was investigated. The results from this study demonstrate that DES is a promising solvent for wood delignification and the production of a new source of lignin with promising potential applications.

Introduction

Lignin is the most abundant renewable aromatic polymer on earth, present in all high plants. Developing effective means to remove lignin from plant biomass and identifying value propositions for lignin products have been pursued for nearly a century by the papermaking industry.[1] Recent efforts towards commercialization of the lignocellulosic biorefinery have injected new enthusiasm into the field of lignin valorization.[2]

Being an integral part of the plant fiber cell wall, lignin cannot be readily separated in a high quantity and high purity without significant alterations of its native structural properties.[3] Current research and development efforts have been primarily focused on the valorization of lignin streams produced from papermaking and biomass refinery processes.[2g,4] The physical and chemical properties of these lignin streams change indiscriminately depending on the process conditions, adding further structural complexity to this already heterogeneous polymer. These lignin products are often contaminated with a significant amount of residual carbohydrates (*e.g.*, diluted acid lignin, steam explosion lignin, mechanical milling lignin) or process chemicals (*e.g.*, lignosulfonates, kraft lignin).

These problems have created enormous challenges, preventing the large scale high value application of this "processed lignin". Native lignin or proto-lignin has been regarded as having great promise as a high value chemical precursor.[2d,e,5] However, due to the lack of practical means of extracting pure native lignin from wood in meaningful quantities, there has been no opportunity to prove the true value of native lignin. Identifying a new method enabling the extraction of a large quantity of lignin from woody biomass with high purity and limited chemical modifications will provide a breakthrough pathway to realize the high value proposition of lignin.

The recent discovery and application of the deep eutectic solvent (DES) provided new insight toward its potential application for lignin extraction.[6] DES is a mixture of two or more chemicals acting as either hydrogen-bond donors (HBD) or hydrogen-bond acceptors (HBA). The results from a few recent studies suggest that DES may have a unique capacity towards dissolving lignin from plant materials.[6b,7] Based on the principles of HBD and HBA interaction, DES may provide a mild acid–base catalysis mechanism which will instigate the controlled cleavage of labile ether linkages among phenylpropane units and thus lead to lignin depolymerization and/or separ-

[a]*Voiland School of Chemical Engineering & Bioengineering Bioproducts, Science & Engineering Laboratory, Washington State University, 2710 Crimson Way, Richland, WA 99354, USA. E-mail: x.zhang@wsu.edu*
[b]*Chemical & Biological Processes Development Group, Pacific Northwest National Laboratory, Richland, WA 99352, USA*
[c]*Composite Materials & Engineering Center, Washington State University, WSU Research Park, Pullman, WA 99164, USA*
†Electronic supplementary information (ESI) available. See DOI: 10.1039/c6gc01007e
‡These two authors contributed equally to this work.

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ation from the biomass. This chemistry may generate a low molecular weight lignin product while maintaining most of the properties and activity of native lignin, and thus has significant potential for industry applications. So far, there has not been a detailed investigation into applying DES for extracting lignin from woody biomass. This study aims to investigate and demonstrate this new lignin extraction method, and characterize the resulting DES extracted lignin product.

Results and discussion

Deconstruction of woody biomass with DES

Douglas fir (D. fir) and poplar were selected to represent the predominant softwood and hardwood species in North America. The chemical composition of samples (wt%) was determined following standard procedures and is presented in Table 1.[8]

Four DES mixtures were prepared using ChCl as HBA with four HBD: acetic acid (ChCl-Aa), lactic acid (ChCl-Lac), levulinic (ChCl-Lev) acid and glycerol (ChCl-Gly). Treatment of woody biomass by each DES was conducted at four different temperatures: 90 °C, 120 °C, 145 °C and 180 °C, with different treatment times. The four treatment conditions are labeled A, B, C, and D respectively (Table 2) and represented four levels of treatment severities based on the *H* factor,[9] a parameter combining temperature and time for controlling lignin removal in the chemical pulping process. A and B are mild extraction conditions.[10] C corresponds to typical chemical pulping conditions applied on hardwood (*H* factor ~ 600) and softwood (*H* factor ~ 1000).[9] D represents typical softwood pretreatment severity.[8b,11]

The solids remaining (solid yield) after DES treatment of poplar and D. fir is presented in Fig. 1. It was shown that the amount of woody components extracted by DES increased with increase in treatment severity. However, hardly any wood components were removed by ChCl-Gly under all conditions. The removal of specific components – cellulose, hemicellulose, and lignin – by ChCl-Aa, ChCl-Lac and ChCl-Lev under each treatment condition was then investigated. As shown in Fig. 2, most treatment conditions caused little cellulose removal from poplar. Only ChCl-Aa and ChCl-Lac treatments at 180 °C led to an appreciable amount of cellulose reduction at 9.4 g and 9.0 g per 100 g of biomass, respectively (Fig. 2D). The removal of both hemicellulose and lignin from poplar by DES treatment was significant and increased along with increasing treatment severity, reaching a maximum at 145 °C (*H* factor 629).

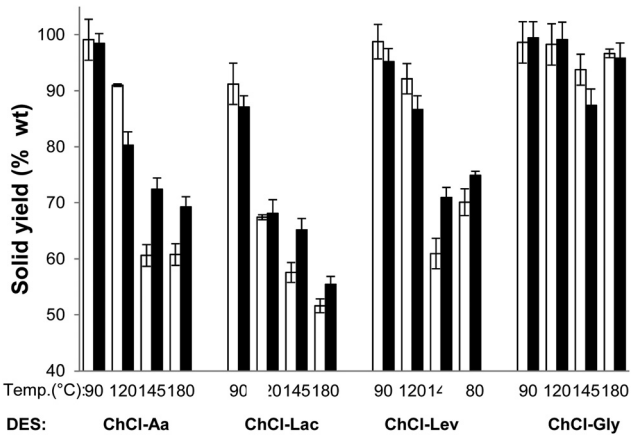


Fig. 1 Solid yields for poplar (□, white bar) and D. fir (■, black bar) as a function of reaction temperature and DES treatment.

Among the three major wood components, hemicellulose is the most susceptible to chemical and thermal degradation and is thus expected to be removed by DES treatment due to hydrolysis.[3a,12] The released sugars were present as a mixture of oligosaccharides and monosaccharides. An appreciable amount of acetic acid was also present, however, little sugar degradation compounds, such as furfural, were detected. The ability of DES extraction of a large amount of lignin from wood was not known previously. At 90 °C, approximately 18.1%, 25.2% and 21.3% of lignin was extracted by ChCl-Aa, ChCl-Lac and ChCl-Lev, respectively. Increasing temperature to 120 °C led to a moderate increase in lignin extraction obtained from ChCl-Aa and ChCl-Lev, but a significant jump in lignin extraction yield to 72.1% by ChCl-Lac. After increasing the temperature to 145 °C, all three DES extracted more than 70% of lignin initially present in poplar with ChCl-Lac exhibiting the highest lignin extraction yield of 78.5% (20.6 g of lignin from 100 g of wood with an initial lignin content of 26.3%). However, a further increase in temperature to 180 °C resulted in a reduction of lignin extraction yield from all treatments. Up to 74.8% of hemicellulose was also removed from poplar along with lignin extraction.

DES treatment of D. fir (Fig. 3) also showed limited removal of cellulose until the treatment temperature reached 180 °C. There was an appreciable amount of hemicellulose released

Table 1 Wood component analysis of D. fir and poplar

Composition	D. fir (wt%)	Poplar (wt%)
Glucan	48.0 ± 0.9	39.7 ± 1.0
Mannan	11.4 ± 0.8	2.7 ± 0.1
Lignin	27.7 ± 0.8	26.3 ± 0.3
Xylan	2.7 ± 0.1	12.4 ± 0.5
Galactan	2.2 ± 0.1	0.4 ± 0.1
Arabinan	0.5 ± 0.1	0.3 ± 0.1

Table 2 DES treatment conditions tested and its equivalent *H* factor

Treatment label[a]	Temperature [°C]	Reaction time [h]	*H* factor
A	90	6	1.8
B	120	3	27
C	145	6[b]	629[b]
		9[c]	943[c]
D	180	0.5	1038

[a] Treatment label corresponds with the used nomenclature in Fig. 2 and 3. [b] Treatment conditions for poplar. [c] Treatment conditions for D. fir.

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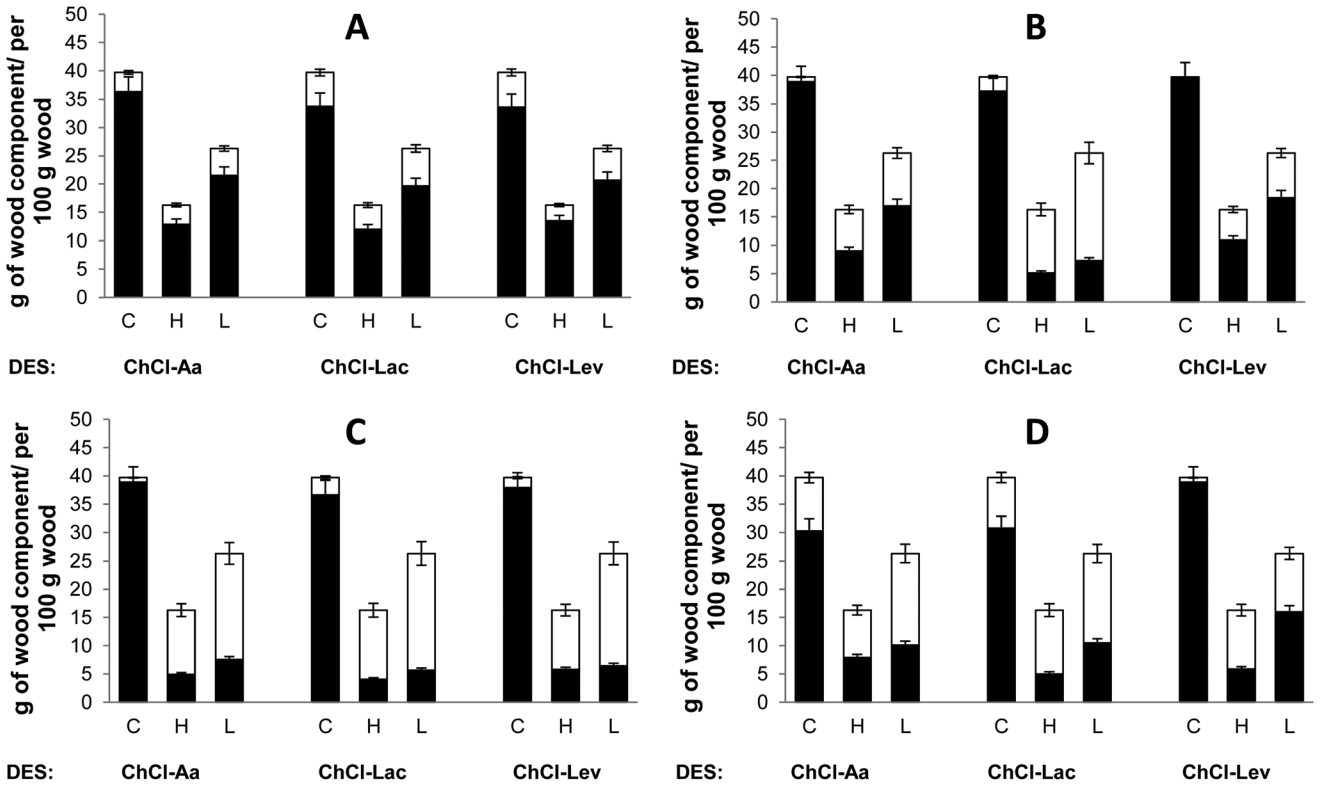


Fig. 2 Poplar wood component distribution after DES treatment as DES solubilized (□, white bar) and insoluble solid (■, black bar) for the main biomass constituents (C: cellulose, H: hemicellulose, L: lignin) as a function of DES type and reaction temperature: (A) 90 °C, (B) 120 °C, (C) 145 °C and (D) 180 °C.

during DES treatment. The amount of lignin extracted from D. fir also increased with increasing temperature up to 145 °C. Maximum lignin extraction yields were observed at 145 °C from all DES (Fig. 3C). Again ChCl-Lac appeared to be the best DES for lignin extraction with 58.2% lignin extraction yield (16.1 g of lignin obtained from 100 g of D. fir with an initial lignin content of 27.7%). A drop in lignin extraction yield at 180 °C was also observed with DES treatment of D. fir softwood.

These results suggest that the DES has the unique capability to selectively extract lignin and hemicellulose from woody biomass while preserving the cellulose. This provides a new opportunity for the simultaneous production of high value lignin and cellulose fibers for papermaking.

Recovery and purification of lignin

After DES treatment, anhydrous ethanol was added to the treated wood to separate the remaining solids from the DES and dissolved organics (ESI Fig. S2†). The ethanol soluble fraction containing dissolved lignin and hemicellulose was collected after filtration. Adding DI water to this filtrate precipitated lignin while the hemicellulose remained in solution. The amount of precipitated lignin was calculated and is shown in Fig. 4. The amount of precipitated lignin ranged from 9.5 g to 46.4 g based on 100 g of initial wood material, depending on the HBD and temperature conditions. Similar to the trend of lignin solubilization during DES treatment (Fig. 2 and 3), the amount of precipitated lignin increased with treatment temperature from 90 °C to 145 °C and then decreased at 180 °C. The maximum lignin recovery from the ChCl-Lac treatment of poplar and D. fir was also confirmed at 145 °C.

The initial lignin content of poplar and D. fir are 26.3% and 27.7% (or 26.3 g per 100 g and 27.7 g per 100 g wood), respectively. Therefore, higher lignin recovery yields above these values (Fig. 4) suggested the presence of impurities in the precipitated lignin. A compositional analysis of the precipitated lignin showed that the purities of the lignin obtained from 145 °C ChCL-Lac treatment of poplar and D. fir were less than 60%. There is a significant amount of DES (>30%) detected based on lactic acid content. The amount of hemicellulose content in the precipitated lignin was insignificant (less than 5%).

It appears that although the addition of DI water helped separate lignin from hemicellulose, it caused co-precipitation of DES with lignin. To reduce the association between DES and lignin during precipitation, a number of solvents and solvent/water mixtures were tested. A summary of the results is shown in Table S1 (ESI†). We found that water and ethanol mixtures with 10–40% v/v ethanol content can significantly reduce DES co-precipitation with lignin. The composition of precipitation lignin products obtained from these mixtures instead of DI water is shown in Table 3. The purities of lignin

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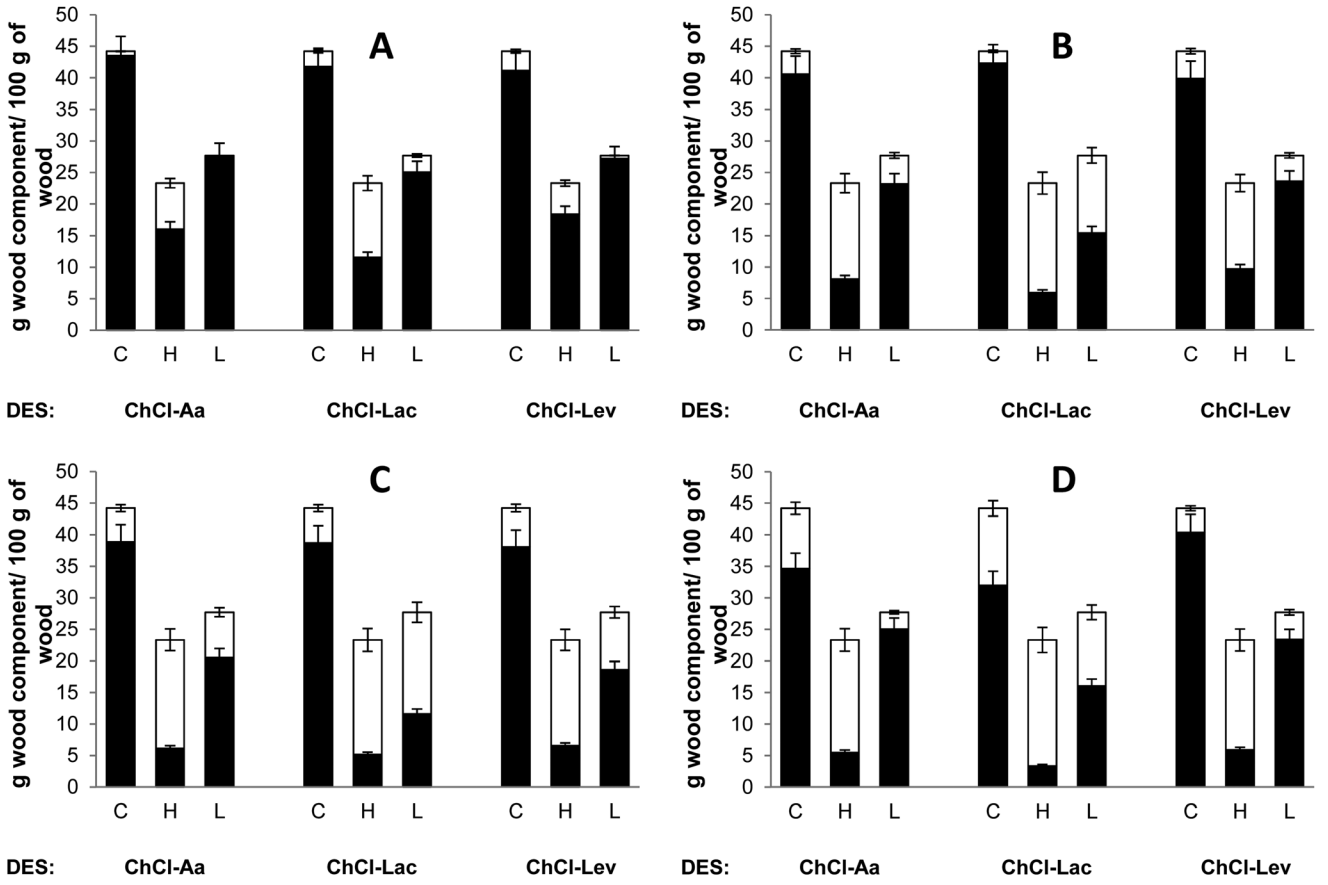


Fig. 3 D. fir wood component distribution after DES treatment as DES solubilized (□, white bar) and insoluble solid (■, black bar) for the main biomass constituents (C: cellulose, H: hemicellulose, L: lignin) as a function of DES type and reaction temperature: (A) 90 °C, (B) 120 °C, (C) 145 °C and (D) 180 °C.

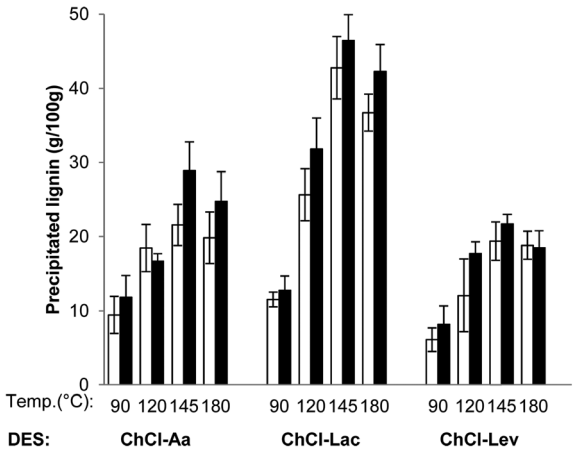


Fig. 4 The amount of precipitated lignin (g lignin per g biomass) from spent-DES after water addition for poplar (□, white bar) and D. fir (■, black bar) as a function of reaction temperature and DES treatment.

obtained from poplar and D. fir can reach approximately 66% and 88%, respectively. Only a small amount of hemicellulose contamination was present in both lignin products. However, a considerable amount of DES still remained: 8.1% w/w in D. fir lignin and 26.7% DES in poplar lignin. This result appears to suggest that DES has an affinity to lignin. Forming a DES between ChCl and phenols has been reported previously.[13] It is reasonable to infer that the choline chloride in the initial DES can bind with phenolic groups on lignin. It was interesting to find that the binding strengths between DES with softwood lignin, predominantly guaiacyl (G) type, and hardwood lignin, a mixture of guaiacyl and syringyl (GS), differ significantly.

The amount of residual DES in D. fir lignin can be minimized by two additional cycles of washing with a 9 : 1 water/ethanol solution. A final lignin product, DESL, with a purity of 95.4% was subsequently obtained with a negligible amount of DES (Table 3 and ESI, Fig. S4†). This DESL is among the highest purities of lignin that has been produced by woody biomass.

Characterization of DES lignin

To demonstrate the unique characteristics of DESL, quantitative ^{13}C NMR, HSQC NMR, and gel permeation chromatography (GPC) analyses were conducted. Milled wood lignin (MWL) extracted from the same D. fir was also prepared and used as a

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Table 3 Precipitation and purity of precipitated lignin with water/ethanol mixtures

	Carbohydrate [%]		AIL [%]		ASL [%]		Residue DES [%]		Lignin purity[c] [%]	
Water : ethanol	P[b]	D[b]	P	D	P	D	P	D	P	D
9 : 1	4.1	1.3	62.5	85.1	3.7	3.3	26.7	8.1	66.2	88.4
8 : 2	2.9	2.0	55.5	84.9	2.3	2.7	36.5	7.5	57.8	87.7
7 : 3	2.2	2.5	40.5	87.1	2.3	1.5	49.5	10.8	42.8	88.6
6 : 4	1.5	2.7	35.5	83.5	2.6	1.3	54.2	13.9	38.1	84.9

	Carbohydrate [%]	Residue Lac [%]	AIL [%]	ASL [%]	Ash [%]	Unidentified [%]	Lignin purity [%]
DESL[a]	Negligible	Negligible	93.1	2.4	0.4	~4.2	95.4

[a] DESL: further purified lignin by repeated washing with 9 : 1 water/ethanol mixture. [b] P: poplar; D: Douglas fir. [c] Lignin purity[%] is calculated from sum up of AIL[%] and ASL[%] of each sample respectively.

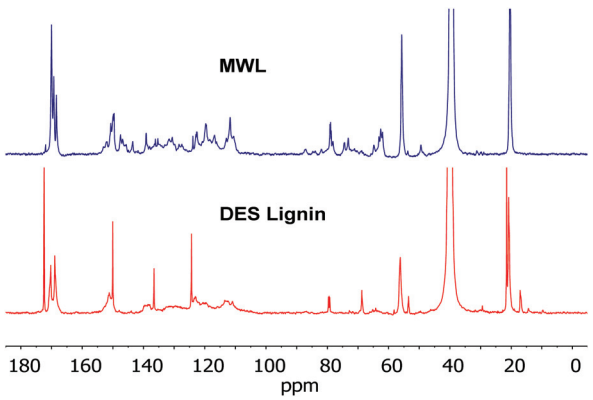


Fig. 5 ^{13}C NMR spectra (ppm) of acetylated DESL and MWL from D. fir.

reference for comparison with DESL in these analyses. The ^{13}C NMR spectra of acetylated lignin samples from DESL and MWL are shown in Fig. 5 with a summary of the major structural parameters presented in Table 4.

The lack of signals between 102–90 ppm confirms the absence of carbohydrates in DESL. It was very intriguing to find that the peak signals between 90–72 ppm were absent suggesting a negligible amount of ether linkages present in this pure DES lignin product. Another striking observation is the higher phenolic hydroxyl groups (168.6–166 ppm) in DESL compared to MWL and other typical softwood lignins,[5,14] which confirms the cleavage of ether linkages. The higher phenolic hydroxyl group content also contributed to the binding of DES with extracted lignin. The amounts of primary (172–169.6 ppm) and secondary (169.6–168.6 ppm) aliphatic hydroxyl groups in DESL were, however slightly lower than that of MWL. The absence of the β-O-4 ether linkage (S1) in DESL was also substantiated by HSQC NMR spectra (Fig. 7). HSQC NMR spectra illustrated that carbon–carbon (C–C) bonds such as the β–β (S2) and β-5 (S3) linkages remained the predominant interunit linkages in DESL. It was also interesting to detect an appreciable amount of Hibbert's ketone (HK) in DESL which further verifies the cleavage of β-O-4 linkages by DES. No condensation structures were detected. The molecular weight distribution and peak molecular weight of this DES lignin was then determined by GPC analysis (Fig. 6). Compared to MWL as well as other typical softwood lignins, DESL had a narrower and lower molecular weight distribution ranging from 490–2600 g mol^{-1} with a peak molecular weight of about 890 g mol^{-1}. This provides evidence that DES depolymerizes native lignin by selective cleavage of ether bonds

Table 4 Quantitative ^{13}C NMR analysis of MWL (D. fir lignin) and DESL (ChCl-Lac treated D. fir lignin) structure

Spectral region	Chemical shift (ppm)	Moieties per Ar	
		MWL	DESL
CH_3O	57–54	0.97	0.90
C_{Ar}–H	125–103	2.75	3.09
C_{Ar}–C	141–125	1.66	1.69
C_{Ar}–O	160–141	1.59	1.22
Phenolic OH	171–174	0.05	0.49
Aliphatic OH	171–168.5	1.62	1.54
Aliphatic side chain	40–20	1.83	1.97
C-γ in β-5 and β-O-4 with C=O	64–62	0.65	0.09
Cα in β-O-4	71–78	0.57	ND
Cβ in β-O-4	83–78	0.42	ND

ND: not detected.

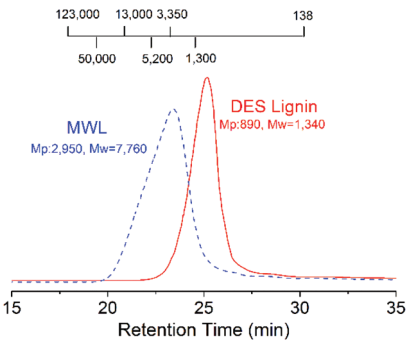


Fig. 6 Molecular weight distribution of acetylated purified DESL from D. fir compared with Milled Wood Lignin (MWL) from D. fir.

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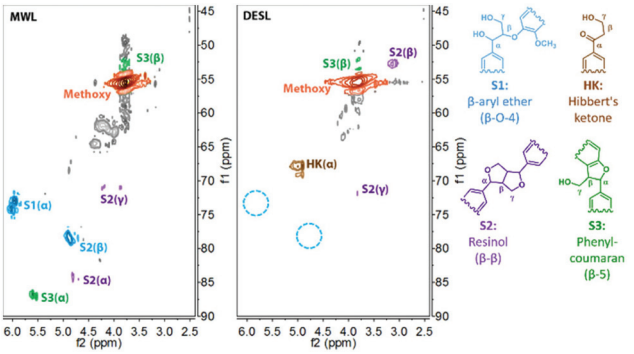
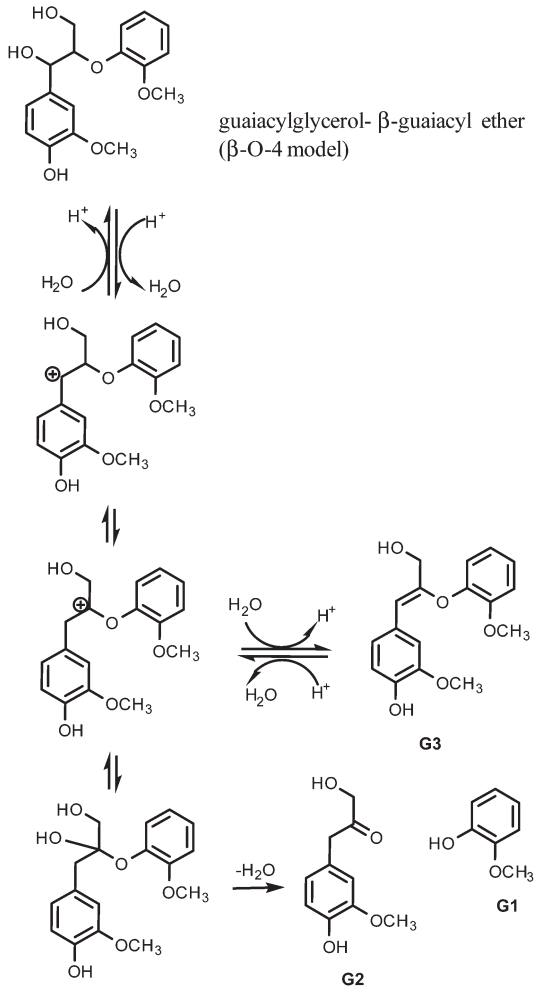


Fig. 7 HSQC NMR spectra (ppm) of acetylated DESL and MWL from D. fir.

without subsequent condensation reactions. This characteristic is distinct from other lignin depolymerization methods.[1a,b] It is conceivable that as ether bonds are more susceptible to disruption than C–C bonds, a reduction or elimination of ether bonds in lignin will likely afford DESL with better polymer stability. It is evident that DESL possesses unique structural properties that are distinctive from all existing prepared lignin, with a great potential for high value applications.

Potential mechanism of DES extraction of lignin from wood

The elucidation of the main characteristics of DESL provides insight to the potential mechanism involved in DES extraction of lignin from wood. To better understand this unexplored reaction mechanism, guaiacylglycerol-β-guaiacyl ether (GBG), a β-O-4 linkage lignin dimeric model compound, was used and treated by ChCl-Lac DES under 145 °C. The products collected during the reaction were analyzed by GC-MS (ESI Fig. S5†). The GBG was found to be completely converted resulting in almost stoichiometric production of guaiacol (G1) and a Hibbert's ketone (HK)-type compound (G2) with G3 detected as an intermediate (Scheme 1). This reaction mechanism is similar to lignin acidolysis catalyzed by HCl.[15] However, the precise roles DES constituents play in the lignin depolymerization require further investigation. It is also very interesting to observe, that compared to acidolysis, very little by-products or phenolic recondensation products were observed during DES treatment. DES treatment of hydroxymatairesinol (HMR), a β–β (C–C bond) linked dimeric model compound, was also conducted, no fragmentation products were detected even after 3 hours of reaction. These results confirm that DES has the ability to selectively cleave ether bonds without affecting the C–C linkages in lignin. The amount of ether linkages represents 40–60% of the total bonds among phenylpropane units in wood lignin.[3c] The cleavage of ether bonds is the main mechanism leading to lignin depolymerization and facilitates lignin extraction from wood. Hardwood lignin has a higher amount of ether bond linkages than softwood lignin. The higher lignin extraction yield from poplar compared to D. fir supported the fact that ether bond cleavage played an important role in DES extraction of lignin. It is possible that poplar DESL has a high



Scheme 1 Main mechanisms of the β-O-4 linkage cleavage reaction during DES treatment of lignin based on study of the model compound guaiacylglycerol-β-guaiacyl ether.

phenolic hydroxyl content which can result in its high affinity with residual DES. Selective cleavage of ether linkages in lignin is also the main delignification mechanism employed by Kraft pulping chemistry.[1b,3c] However, a key difference between DES treatment and Kraft pulping is the absence of recondensation of lignin fragments, likely due to the mild treatment conditions employed.[1b,16]

While the depolymerization of lignin facilitated its separation from wood, it was also probable that the solvent properties of DES played another important role as the lignin extraction yields differed significantly among the different DES mixtures. The efficiency of DES extraction of lignin may be associated with the Kamlet–Taft polarity parameters which have been previously applied to interpret the relationship between solvent properties and their abilities to extract cellulose and lignin.[17] This empirical model characterizes solvents by three parameters: H-bond donating ability or acidity α, H-bond accepting ability or basicity β, and polarity and polarizability π^*.[18] Several studies have suggested that a solvent with

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both high β and π^* can dissolve lignin.[17] DES typically has a β value between 0.6–0.8 and a π^* of around 1.0, which is similar to those of DMSO and pyridine.[18,19] We determined the solubility of lignin obtained from ChCl-Lac extraction by a range of solvents with different Kamlet–Taft polarity values (ESI Table S2†). It was found that dioxane, DMSO and pyridine can dissolve a significant amount of DESL and the solubility of DESL decreased in solvents with lower β and π^* values. These results suggest that DES polarity and basicity indeed played a significant role in lignin extraction. However, there is no prior knowledge of specific interactions between DES and lignin molecule(s) governed by these parameters. A moderately high temperature of 145 °C was unlikely to be able to breakdown all ether linkages in lignin.[1b,3b] However it is perhaps sufficient to degrade and remove hemicellulose and enhance the access of DES to lignin for extraction. A clear understanding of these interactions contributing to a comprehensive mechanism of DES extraction is essential for demonstrating the potential of DES for biomass conversion applications.

Conclusion

In conclusion, this work demonstrated a novel method of applying Deep Eutectic Solvents (DES) to extract high purity lignin (up to 95% from D. fir) from woody biomass in high yield (up to 78% from poplar and 58% from D. fir). The resulting lignin product, DESL, has several distinctive characteristics: lower and narrowly distributed molecular weight compared to MWL, and the lack of ether linkages, representing a new source of lignin. The mechanism involved in DES depolymerization of lignin from wood was investigated suggesting that both DES-catalyzed ether bond cleavage and solvent effects are important to facilitate lignin extraction from wood. The findings from this study may provide a breakthrough toward truly realizing the high value potential of lignin.

Experimental

General procedure for DES preparation

DES preparation was carried out by mixing both starting materials in the solid state, followed by melting them at 65 °C for 2 h in a vacuum oven as per a procedure reported elsewhere.[7] The solid mixtures were stirred periodically during this time until a homogeneous and transparent liquid was obtained. Once the liquid was formed with no evidence of solid particles, the mixture was cooled down in a desiccator to room temperature to avoid moisture absorption (ESI Fig. S1†).

Treatment of lignocellulosic biomass with DES

The procedure for biomass and DES separation after treatment includes three main steps (ESI Fig. S2†). Prior to DES treatment, the samples were air-dried and milled to a specific size (mesh −40/+60). The treatments were carried out using a biomass concentration of 10 wt% (0.6 g of dry biomass and 6 g of DES) in reaction vessels.[8b,10]

General procedure for biomass, lignin and DES separation after treatment

After treatment, the solid fractions and DES soluble fraction were first separated by filtration using a glass crucible with ethanol washing (ESI, 1st step, Fig. S2†).[7] DI water was then added to the filtrate in the second step to precipitate lignin. Lignin was collected after centrifugation and washing with additional water (water/ethanol mixture) to increase purity (2nd step, Fig. S2†). The ethanol was removed from the supernatant by vacuum evaporation at 65 °C and the ensuing DES/water solution was then dried in the oven at 105 °C for 4 hours to remove water and regenerated DES (3rd step). The composition analyses of solid fraction and collected lignin were conducted following standard NREL procedures.[8a,20] The solid fractions obtained from the different DES treatments and subsequent washing are shown in Fig. S3.†

Lignin purity determination

Lignin purity was calculated based on the total amount of acid soluble lignin (ASL) and acid insoluble lignin (AIL) in samples determined according to standard procedures.[20] In addition, the content of impurities including sugars and organic acids (acetic acid, lactic acid and levulinic acid) associated with residual DES was determined by HPLC using a Biorad Aminex H column according to previous procedures.[21] The amount of residual DES in lignin was then estimated based on the ratios between respective organic acids and ChCl.

^{13}C/HSQC NMR and molecular weight distribution analysis of DES lignin

D. fir milled wood lignin (MWL) preparation. Dioxane-HCl extracted MWL was prepared from the same D. fir softwood following the previous procedure[22] with slight modification and used as a reference for DESL lignin characterization. The D. fir wood mill was ground in a laboratory vibration ball-milled for forty-eight hours. The ground wood mill was then subjected to cellulase enzyme (Ctec 2 from Novozymes, North America) hydrolysis for another forty-eight hours to remove the residual carbohydrates. The solid residue after enzymatic hydrolysis was then washed with acetate buffer and freeze dried. The lignin in the freeze dried sample was extracted by dioxane/water/HCl for 24 hours[23] and collected as MWL.

Structural characterization of purified D. fir DES lignin (DESL) and D. fir milled wood lignin (MWL) by the ^{13}C NMR was conducted on a Bruker 500 MHz NMR spectrometer with a 90° pulse width, a 1.4 s acquisition time, and a 1.7 s relaxation delay following standard procedures for lignin analysis.[24] Both acetylated and native lignin samples were analyzed using DMSO-d6 as a solvent (100 mg sample per 0.6 ml DMSO-d6). A small amount of relaxation agent ($Cr(acac)_3$, 2 mg) was added during sample preparation to facilitate the relaxation of the magnetization.[25] The 2D HSQC NMR spectra of the acetylated DESL and MWL were recorded on the same Bruker 500 MHz

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NMR spectrometer at 25 °C using the Q-CAHSQC pulse program. Matrices of 2048 data points for the ^{1}H-dimension and 1024 data points for the ^{13}C-dimension were collected from 13 to −1 ppm and from 160 to 0 ppm for the ^{1}H and ^{13}C dimensions, respectively. Relaxation delay was set at 6 s. The lignins were dissolved in DMSO-d6 and chemical shifts were referenced to the solvent signal (2.50/39.5 ppm).

The molecular weight distribution of DESL and MWL was determined by gel permeation chromatography (GPC).[26] The lignin samples were acetylated in pyridine and an acetic anhydride mixture prior to analysis. Approximately 2 mg of sample was dissolved in tetrahydrofuran and then filtered through a 0.45 μm filter. A Perkin-Elmer High performance liquid chromatograph (HPLC) equipped with a diode array detector (DAD) was used to conduct the gel permeation chromatography (GPC) analysis of the lignin molecular weight distribution. A series of three Agilent ZORBAX PSM columns (60S, 300S and 1000S) was used to elute different lignin molecular weight fractions by tetrahydrofuran at a flow rate of 0.6 ml per minute for 60 minutes. Polystyrene standards (Alfa Aesar) with molecular weights ranging from 1300 to 123 000 g mol^{-1} as well as veratrol (138 g mol^{-1}) were used to calibrate the molecular weight based on retention time.

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